UNIVERSITY EXAMINATIONS: 2010/2011

FIRST YEAR STAGE EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

BIT 1101: LINEAR ALGEBRA

DATE: AUGUST 2011 **TIME: 2 HOURS**

INSTRUCTIONS: Answer question ONE and any other TWO questions

Question One

a) Define the following terms as used in Algebra: (9 Marks)

i) Eet

ii) Product set

iii) Tautology

iv) Power set

v) Proposition

vi) Conjuction

vii) Contradiction

viii) Argument

ix) Fallacy

b) Draw a Venn diagram and shade the region corresponding to the indicated set. (3 Marks)

$A^c \cap (B \cup C)$.

c) Prove by induction that $1^2+2^2+3^2+\ldots+n^2= \frac{n(n+1)(2n+1)}{6}$ (5 Marks)

d) In a survey of 60 people, it was found that 25 read Newsweek, 26 read Time and 23 read fortune. Also 11read both Newsweek and Time, 9 read Newsweek and Fortune, 8 read both Time and Fortune, and 3 read all three magazines. Find the number of people who read: (8 Marks)

i) Only Newsweek

ii) Only Time

iii) Only Fortune

iv) Newsweek and Time, but not fortune

v) Only one of the magazines

vi) None of the magazines.

e) Show that "p implies q and q implies p" is logically equivalent to the bicondittional "p if and only if q". (5 Marks)

Question Two

a) Given that f(x) = 3x + 1 and g(x) = $\frac{x}{4}$, show that $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$. (4 Marks)

b) Translate into symbolic form and test the validity of the argument:

c) If l work, l cannot study. Either l study or l pass mathematics. I worked. Therefore l passed mathematics. (6 Marks)

d) Given that A = $\{a, b, c, d\}$. Find the power set of A. (3 Marks)

e) State the principle of duality in set theory. (2 Marks)

f) Use crammers rule to solve the system of equations (5 Marks)

$$2x + y + z = -2$$
$$x - 3y - z = -10$$
$$-5x + 2y + 3z = 13$$

Question Three

a) Express the specification "The automated reply cannot be sent when the file system is full" using logical connectives. (4 Marks)

b) Determine whether these system specifications are consistent:

"The diagnostic message is stored in the buffer or it is retransmitted"

"The diagnostic message is not stored in the buffer"

"If the diagnostic message is stored in the buffer, then it is retransmitted". (5 Marks)

c) Use predicates and quantifiers to express the system specifications " Every mail message larger than one megabyte will be compressed” and “ if a user is active, at least one network link will be available” (3 Marks)

d) Consider these statements of which the first three are premises and the fourth is a valid conclusion.

“All hummingbirds are richly covered”

“No large birds live on honey”

“Birds that do not live on honey are dull in color”

“Hummingbirds are small”.

Let P(x), Q(x), R(x) and S(x) be the statements “x is a hummingbird”,”x is large”, “x lives on honey”, “x is richly colored”, respectively. Assuming that the domain consists of all birds, express the statements in the argument using the quantifiers P(x), Q(x), R(x) and S(x). (8Marks)

Question Four

a) Find a matrix A such that $\begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix} A = \begin{pmatrix} 3 & 0 \\ 1 & 2 \end{pmatrix}$ (6 Marks)

b) Solve the following system of equations

$$\begin{aligned} 7x - 8y + 5z &= -6 \\ -4x + 5y - z &= 5 \\ x - y + z &= 0 \end{aligned}$$

Using

i) Substitution and elimination method (5 Marks)

ii) Inverse method (9 Marks)

Question Five

a) How many different six figure phone numbers are possible if the digits 0 to 9 are allowed except that the first digit must be 7,8 or 9. (2 Marks)

b) Let A = $\{1,2\,3\}$, B = $\{a,b\,c\}$ and C = $\{x,y,z\}$. Consider the following relations R and S from A to B and from B to C respectively:

R = $\{(1,b),(2,a),(2,c)\}$ and S = $\{(a,y),(b,x),(c,y),(c,z)\}$. Find (6 Marks)

i) The composition relation RoS

ii) The matrices M_R, M_S and M_{RoS} of the respective relations R, S and RoS and compare M_{RoS} to the product $M_R M_S$.

c) Consider the following relations on the set A = $\{1,2\,3\}$: (12 Marks)

R = $\{(1,1),(1,2),(1,3),(3,3)\}$

S = $\{(1,1),(1,2),(2,1,),(2,2)(3,3)\}$

T = $\{(1,1),(1,2),(2,2)(2,3)\}$

d) Determine whether or not each of the above relations on A is

i) Reflexive

ii) Symmetric

iii) Transitive

iv) Antisymmetric

e) Give reasons for your answer